

Homework 2

Due Wednesday, September 13

1. A bead (mass m) glides without friction on a metal rod that makes an angle α with the vertical and rotates about an end (which is placed at the origin) with angular velocity ω . In this problem, we will neglect gravity.
 - (a) Choose the distance r of the bead from the origin as the generalized coordinate. Obtain the Lagrangian of the bead $L(r, \dot{r}, t)$ and derive the Euler-Lagrange (EL) equations of motion.
 - (b) If the bead starts at $r(0) = r_0$ at $t = 0$ with zero velocity, what will be $r(t)$ at any future time t ?
 - (c) Suppose we want to determine the reaction force on the bead. One way to do that is to extend the set of generalized coordinates and impose the constraints. Suppose we extend the set to

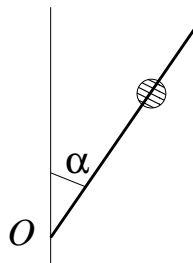
$$q = (r, \theta),$$

where θ is the angle the rod makes with the vertical, and impose the constraint

$$\theta(t) = \alpha = \text{const.}$$

Obtain the Lagrangian $L(q, \dot{q}, t)$ and use the method of Lagrange multipliers to derive the EL equations of motion for $r(t)$ and $\theta(t)$.

- (d) Observe that the method described in (c) allows you to determine the component of the reaction force in the $\hat{\theta}$ direction. Find that component for the solution obtained in part (b).



2. Consider the system (a disk rolling on a parabola) described in Problem 25 to Chap. 2 of the textbook. In this case, the no sliding constraint is holonomic, meaning that the

angle ϕ by which the disk has rotated about its center of mass can be expressed through a generalized coordinate q describing translational motion. Let us choose $q = x$, where x is the x coordinate of the point of contact.

(a) Express ϕ as a function of x . (Please note that ϕ is the angle defined above and **not** the angle that the tangent to the parabola makes with the horizontal.)

(b) Express coordinates of the center of mass x_{cm} and y_{cm} through $q = x$. Express the velocities $\dot{x}_{cm}$, $\dot{y}_{cm}$ through $\dot{q}$.

(c) Consider the motion as a composition of translation and rotation about the center of mass and use results of parts (a), (b) to obtain the Lagrangian $L(q, \dot{q})$.